MScFE 652 PORTFOLIO MANAGEMENT

Group Work Project # 2

See grading rubric here

Scenario

John Wolf is the senior portfolio manager of Smart Money Inc., a wealth management company specializing in using different advanced methodologies to allocate assets to its clients' portfolios. You have been hired as his quantitative analyst to perform the actual optimizations for the company.

Your boss requested you considering the following 7 assets as suitable for his clients:

{AAPL, NVDA, TSLA, XOM, REGN, LLY, JPM}.

Furthermore, Mr. Wolf instructed you to use:

- Jan 1, 2023, JUne 30, 2025 as the appropriate period for the task;
- Use either weekly or daily data to estimate the parameters of your models.

Mr. Wolf prepared the following list of tasks that you are expected to complete

Part 1

Step 0: Prepare a quick summary of your data (mean, volatility, skewness, kurtosis and a correlation heatmap). You can provide any short comment that you consider relevant. Be professional and brief.

Step 1: Determine the optimal weights for the assets of the portfolio assuming that short selling is not allowed.

Step 2: In order to have a more diversified portfolio, Mr. Wolf wants to see what the optimal portfolio turns out to be if short selling is not allowed and, in addition, no assets can represent more than 20% of the portfolio.

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Step 3: For some more 'aggressive' clients who are fine with using some leverage (i.e. short selling), Mr. Wolf asked you to determine the optimal portfolio when short selling is allowed.

Step 4: Present the weights of the optimal portfolios in Step 1, 2, and 3, in a table and write some comments that you deem appropriate to describe the findings. Make sure to explain the possible advantages and disadvantages of each approach.

Important: Complete Steps 1, 2, and 3 using optimization (**NOTE: do not use weights simulation; you MUST use true optimization. No credit will be given for using simulations.**)

Part 2

Mr Wolf has a client, Dr. Tobias Smollett, who is a curious investor. Dr. Smollett tells Mr. Wolf that in his opinion a "1/N portfolio" is a better approach. Mr. Wolf wants to show Dr. Smollett that this is not the case in general. To prove it, he instructs you to create a simulation to compare the performance of MVO portfolios and 1/N portfolios as follows:

- Determine a period of 3 years, then split it into two parts: one period of 2 years to estimate the parameters and one period of 1 year to test the portfolios.
- Create an investable universe (ideally this should be 2-3000 assets, but for simplicity limit yourselves to 20 assets; further, you may use all equities so that it is easier to find the data, however this is not mandatory).
- Selects 5 assets at random from the 20 assets and do the following:
    - Compute the MVO portfolio (no short selling allowed)
    - Compute the 1/N portfolio (i.e. the portfolio with all weights equal to 0.2).
- Consider a period of 1 year going forward and compute the returns for each of the two portfolios created above.
- Repeat this for at least 5,000 simulations to obtain 2 series of 5,000 returns each, one for the MVO portfolio and one for the 1/N strategy.
- Plot the distributions of returns for the two strategies and compute relevant statistics to show Dr. Smollett.

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Step 5: Write the code to perform the simulations as described above and prepare a 1 page presentation that contains the key results and provides the relevant conclusions from your analysis.

Part 3

Later, Mr. Wolf, being an expert, decides that his clients pay him for his ability to generate portfolios with views and exploit his expertise. In addition, Smart Money Inc. has a team of analysts who are skilled at creating forecasts for several assets.

Step 6: Pretend that you are these analysts and using several sources formulate views for your 7 assets listed in the scenario above. Make sure that you have at least one view per asset. Ideally do not just use absolute views with no uncertainty every time.

Step 7: Compute the weights for the new optimal portfolio using the Black Litterman model.

Part 4

Finally, a different group of clients has asked Mr. Wolf to use the Kelly criterion to create their portfolio. For these clients, you are required to prepare a short presentation that considers the half-Kelly and the double-Kelly strategy and to explain if and when each of the different strategies (Kelly, half-Kelly, double-Kelly) outperformed the others.

Step 8: Prepare this presentation (about 2 pages) that shows the results from the strategies based on Kelly criterion. Make sure that you do not assume that the assets are uncorrelated and absolutely do not use the gambling formula which is not suitable for portfolio applications. **YOU MUST USE the most general formula to compute the weights**. Make sure that you explain the steps briefly. Do not try to summarize the theory. Explain what you did using the results. The comparisons should include both quantitative metrics and/or graphical and tabular visualizations.

Groups of 3 students are responsible for all steps.

Groups of 2 students are responsible for Steps 0, 1, 2, 3, 4, 5 and EITHER Step 6 and Step 7 OR Step 8.

Groups of 1 student are responsible for Steps 0, 1, 2, 4, 5, EITHER Step 6 and 7 OR Step 8.

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Submission requirements and format

READ the [WQU Academic Policy on the Use of AI](#) for a proper use of this tool. Instructors heavily penalize any improper use.

One team member submits on behalf of the entire group the following:

1. **1 PDF document*** Following the suggestions for the different parts/steps above, your paper should consist of anywhere between 10 and 12 pages, including some relevant graphs and/or tables. Note that graphs and tables that are relevant to your paper should appear in the paper itself.
    a. Use the available Report Template and fill out the required information on the first page, then make sure to include the reports in the PDF along with the template.
2. 1 **zipped folder** including:
    a. One (1) Jupyter notebook with the code. Your code MUST contain comments where you explain your procedures and choices.
    b. One (1) pdf or html file version of the Jupyter notebook that contains the code, comments, AND results. The explanation of your results can be contained here and/or in the paper. The explanation of the results would fit better in the paper

** **Use Google Docs to collaborate**. Start by uploading the Report Template provided in the Course Overview. Once your report is completed, click File → Download → PDF Document (.pdf) to obtain the copy for your submission.*

*** **Use Google Colab or GitHub to collaborate** in completing the executable Python program.*

*The PDF files must be uploaded **separately** from the zipped folder that includes any other types of files. This allows Turnitin to generate a similarity report.*

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Rubric

Your instructor will evaluate your group submission for GWP1 using the following rubric:

Quantitative Analysis (open-ended questions) 40 Points	Technical and Non-technical Reports 30 Points	Writing and Formatting 20 Points
The group is able to apply results, formulas, and their knowledge of theory to real-life finance scenarios by doing the following: ● Providing all the necessary information to support their arguments. ● Presenting arguments that reflect group discussion and research. ● Using authoritative references to support a position and provide updated information ● Concluding with practical takeaways for more insightful financial decision-making	Technical Reports contain 3 parts: 1) summary of key results; 2) interpretation of results; and 3) the recommended course of action that can reasonably follow from those results and interpretations. **Note**: Technical reports will include the technicalities of models, such as names, methods of estimation, parameter values, etc. and exclude generalities about the work done. It should NOT include names of Python code that were used.	A submission that looks professional should include: ● The axes, labels, and scales in graphs. ● No significant grammar errors or typos. ● Organized, well-structured, and easy-to-read document. ● Proper citations and bibliography using MLA format.
	Non-technical Reports contain 3 parts: 1) clear explanation of results; 2) the recommended course of action that follows; and 3) the identification of factors that impact each portfolio. **Note**: AVOID all references to model names, algorithms, and unnecessary details. Instead, focus on the investment decision.	

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